Actual output - predicted output

(1) Binary Classification

x	y	z	a	output
DSA_Score	IQ	Project	sleep	
100	110	4	8	1
200	100	5	9	1
300	70	2	8	0

0.6

0.1

0

1

$$\text{output} = w_1 x + w_2 y + w_3 z + w_4 a + b$$

$\hookrightarrow 120 \Rightarrow \boxed{0 \text{ to } 1}$

$\hookrightarrow$ sigmoid

$$\left[\frac{1}{1+e^{-z}}\right]$$

40

160

0.6 $\Rightarrow$ 0.4

$\Downarrow$ Loss fn

0.5

Loss func = 0 - 0.6

$\Rightarrow \boxed{-0.6} \Rightarrow$ update

Predic = 0.9

Pred $\longrightarrow$ 0.4

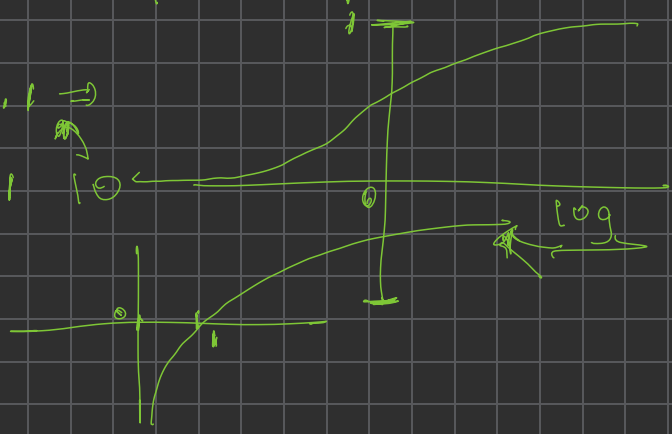
Predic

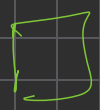
0 = 1

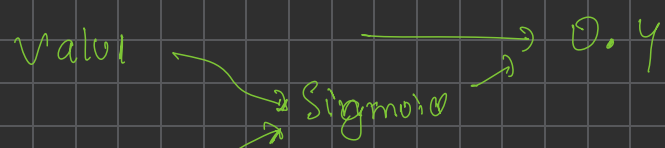
Loss fn 0.1 $\Rightarrow$

Actual output = 1 10

$-\log(p)$







0 — 1

Actual

Predicted output

Actual output = 0.

$\boxed{0 - 0.9}$

$\log(0.9) \Rightarrow -0.2$

$\log(0.8) \Rightarrow -0.5$

$\log(0.4) \Rightarrow -0.6$

$\log(0.2) \Rightarrow -0.8$

Predict

0 - 0.8

0.2	0.4	0.8	0.9
- 0.2	- 0.4	- 0.8	- 0.9

$\log(0.9) \Rightarrow -0.2$

$\log(0.8) = -0.5$

$\log(0.4) = -0.2$

$\log(0.2) \Rightarrow -0.8$

$\log(1-P) \Rightarrow \log(0.1) \Rightarrow -0.9$

$\log(1-0.8) = \log(0.2) \Rightarrow -0.7$

$\log(1-0.4) \Rightarrow \log(0.6) \Rightarrow -0.4$

$\log(1-0.2) = \log(0.8) \Rightarrow -0.2$

Actual output = 1

$-\log(P)$

Actual output = 0

$\log(1-P)$

X	output
0.9	-0.11
0.8	-0.25
0.6	-0.38
0.5	-0.41
0.4	-0.52
0.3	-0.63
0.2	-0.71
0.1	-0.91
⋮	⋮

0.8

$-\log(0.8)$

$\boxed{0.25}$

0.2

$-\log(0.2)$

0.21

$\log(1-0.4)$

$\log(0.6)$

-0.38

0.4

0.9

-0.91

$\log(1-0.9)$

$\log(0.1)$

$\boxed{-0.91}$

$\boxed{\text{Error}}$

$$= -\left(y\log(P) - (1-y)\log(1-P)\right)$$

$\boxed{\text{Derivative}}$

$$= -\left(\log(0.4)\right)$$

$$= -\log(0.4)$$

$-\log(0.1)$

$\boxed{+0.91}$

$\boxed{\text{Initial}}$

$$= -\left(y\log(P) - (1-y)\log(1-P)\right)$$

$$= -\left[y\log(P) + (1-y)\log(1-P)\right]$$

formula

$$\left[W_{new} = W_{old} + 0.01 * \text{Error} * \text{input}\right]$$

$$\left[W_{new} = W_{old} + \eta * \text{error} * x\right]$$

$w_{1new} + 0.01 * (y-p) * x$

$$W_{1old} = W_{1new} + 0.01 * (\text{Error}) * x$$

$$\Rightarrow W_{1new} + 0.01 * (y-p) * x$$

$$\Rightarrow W_{1new} - 0.01 * (p-y) * x$$

error is Actual − Predic

$y - p$

error

$\Downarrow$

(1) $\leftarrow$ $-[y \log p - (1-y) \log(1-p)]$

$$\left[W_{1new} = W_{1old} + \eta * \text{error} * x \right]$$

(2) $-[y \log p + (1-y) \log(1-p)]$ *

$$\Rightarrow \left[W_{1new} = W_{1old} - (p-y) \eta * x \right]$$